STATISTICAL ANALYSIS PIPELINE

Statistical Analysis of Home Ground Advantage in IPL

Deriving team strengths via Ridge regression and testing venue impacts using cluster-robust OLS regressions on ball-by-ball data (2008–2026).

SCOPE & SETUP

Seasons, Teams & Home Venues

Seasons (2008–2026)

Target analysis spans 2008–2026, with train subsets from 2009–2022 and test sets from 2023–2026.

Selected Teams

Focuses on Royal Challengers Bengaluru, Mumbai Indians, Chennai Super Kings, and Kolkata Knight Riders.

Home Venues

Mapped via dictionary lookup (e.g., M Chinnaswamy, Wankhede, Chepauk, Eden Gardens) to classify home status.

FEATURE ENGINEERING

Innings-Level Metrics & Historical History

1. **Batting Performance**
   Runs, Strike Rate, Boundary %, Runs per Wicket, and Dot-ball %.

2. **Bowling Performance**
   Economy, Wicket Rate, Dot-ball %, Boundary Conceded %, and Runs Conceded.

3. **Expanding History**
   Lagged expanding means computed to serve as historical training features for regressions.



MODEL WEIGHTING

Batting & Bowling Model Weights

Batting Weights Dictionary

Runs: 0.30

StrikeRate: 0.25

RunsPerWicket: 0.20

BoundaryPct: 0.15

DotBallPct: 0.10

Bowling Weights Dictionary

Economy: 0.30

WicketRate: 0.25

DotBallPct: 0.20

BoundaryConcededPct: 0.15

RunsConceded: 0.10

AGGREGATION

Season Strength & Rolling History

Rolling Window

For each season, historical data from the prior 3 seasons (`season >= season - 3` and `< season`) is aggregated to compute BattingStrength and BowlingStrength per team.

Min-Max Normalization

Metrics are normalized using min-max scaling, respecting whether higher or lower values are better, before combining via weighted sums.

ECONOMETRIC ANALYSIS

Statsmodels OLS with Clustered SEs

Batting Regression

$$\text{RelativeScore} = \beta_0 + \beta_1 \cdot \text{BattingHome} + \beta_2 \cdot \text{OwnBattingStrength} + \beta_3 \cdot \text{OppBowlingStrength}$$

Standard errors robust to cluster correlation (CRV1) grouped by match ID.

Bowling Regression

$$\text{RelativeRunsConceded} = \beta_0 + \beta_1 \cdot \text{BowlingHome} + \beta_2 \cdot \text{OwnBowlingStrength} + \beta_3 \cdot \text{OppBattingStrength}$$

Quantifies home ground and strength impacts on runs conceded.

PREDICTIVE MODELING

Logistic Regression Match Predictor

Model Specifications

Without Home: Batting & Bowling Strength Differences

With Home: Strength Differences + HomeAdvantage (+1 / 0 / -1)

Evaluation Metrics

Accuracy & Brier Score: Assessed on test set (2023–2026).

Compares predictive performance with and without venue indicators.

OUTPUTS

Generated CSV Artifacts in a .ipynb

Team Strengths

ipl_team_strength_by_season.csv

Regression Results

ipl_batting_regression_results.csv

Predictions

ipl_match_model_comparison.csv

Summary & Conclusion

Strictly modeled on a.ipynb: Combining historical rolling features, Ridge-derived weights, and cluster-robust OLS regressions to quantify home ground advantage in the IPL.